Falling and Forces

Friction is the force that resists the movement of an object as it moves over the surface of a material it is contact with.

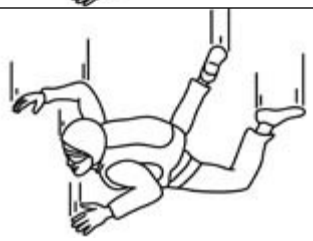
When an object falls there is a friction between the falling object and the air it is falling through. This friction is known as _____ _____.

The greater the surface area an object has the greater the _____ _____.

Falling Ball



Terminal velocity is the highest speed an object reaches as it falls through air. It is reached once the air resistance equals the pull of gravity. The objects keeps falling, but it stops accelerating and falls at a constant speed.

Stage of fall	What is happening
	**Before jumping**. The pulling force of gravity is balanced by the upward force of the plane the pilot is standing in.
	**Speeding up**. Initially the skydiver speeds up because the _____ _____ is low.
	**Terminal Velocity**. As speed increases so does _____ _____. _____ _____ and gravity are b_____. Stops accelerating. The speed of falling remains c_____.
	**Slowing down.** Once the parachute is opened , the force of _____ _____ is greater and the pilot _____ down.